MTH2003

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

January 2021

Differential Equations

Module Leader: Prof. Vadim N. Biktashev

Duration: 2 HOURS (+30 minutes download time)

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN NOTE** examination

USEFUL FORMULAS

These can be used without proof.

Classical Fourier series

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos(nx) + b_n \sin(nx) \right],$$

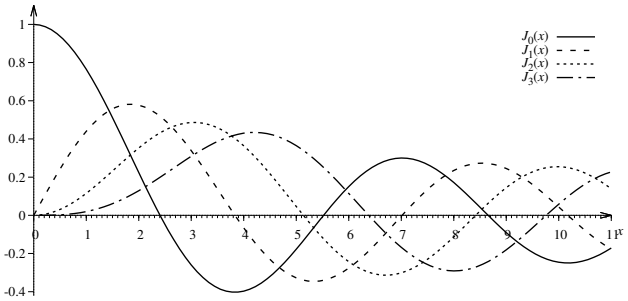
$$a_0 = \int_0^{2\pi} \frac{f(x)}{2\pi}\, \mathrm{d}x,\ a_n = \int_0^{2\pi} \frac{f(x)\cos(nx)}{\pi}\, \mathrm{d}x,\ b_n = \int_0^{2\pi} \frac{f(x)\sin(nx)}{\pi}\, \mathrm{d}x.$$

Bessel equation

$$y'' + \frac{1}{x}y' + \left(1 - \frac{m^2}{x^2}\right) y = 0$$

Solutions that are bounded as $x \to 0$ are known as Bessel functions of the first kind,

$$y(x) = J_m(x).$$



Useful integrals If $j_{m,n}$ is n-th positive zero of $J_m(x)$, then

$$\int_0^1 x\, \left(J_m(j_{m,n}x)\right)^2\, \mathrm{d}x = \frac{\left(J_{m+1}(j_{m,n})\right)^2}{2},$$

$$\int_0^1 x^2\, J_1(j_{1,n}x)\, \mathrm{d}x = -\frac{J_0(j_{1,n})}{j_{1,n}}.$$

SECTION A

1. (a) Find the special and the typical solutions of the ODE

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 8x(x^2-1)(y+1)^{1/2}.$$

Hence find the maximal interval on which the solution satisfying $y(0) = 0$ is unique. (10)

(b) Find the general solution $y(x)$ to the following ordinary differential equation:

$$x^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 6y = -4x^2, \qquad x > 0.$$

Hence find the particular solution such that $y(1) = y(2) = 5$. (10)

(c) Verify that $y = 1/x$ is a solution of the differential equation

$$(2x^2+x)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + (2x+2)\frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 0, \qquad x > 0.$$

Find the general solution to this equation. (10)

(d) Use a trial function of the form $y = \sum_{n=0}^{\infty} a_n x^n$ to find the solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 3x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0.$$

Write the general solution in the form $y = Ay_1(x) + By_2(x)$, where $y_1(0) = 1$, $y_1'(0) = 0$ and $y_2(0) = 0$, $y_2'(0) = 1$. Write down the first three non-zero terms of the power series for $y_1(x)$ and $y_2(x)$. Determine the radius of convergence of the resulting power series. (10)

(e) Consider the eigenfunction problem

$$(1-x^2)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4x\frac{\mathrm{d}y}{\mathrm{d}x} + \lambda y = 0,\ x \in [-1,1]; \qquad |x(-1)| < \infty,\ |x(1)| < \infty.$$

Bring this equation into the Sturm-Liouville form and hence state the orthogonality condition for the eigenfunctions of this problem. Verify that $y_* = 5x^2 - 1$ is an eigenfunction of this problem and find the corresponding eigenvalue λ_*. State how many eigenvalues smaller than λ_* this problem has. (10)

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SECTION B

2. Consider the system of equations

$$\begin{aligned} \frac{\mathrm{d}u}{\mathrm{d}t} &= u - 2v + g_1(t), \\ \frac{\mathrm{d}v}{\mathrm{d}t} &= 4u - 5v + g_2(t), \end{aligned}$$

where $g_1(t)$, $g_2(t)$ are given functions of time.

(a) Find the general solution of this system for $g_1(t) \equiv 0$, $g_2(t) \equiv 0$. (14)

(b) Find the solution of this system for $g_1(t) = 10\cos t$, $g_2(t) \equiv 0$, for the initial conditions $u(0) = 0$, $v(0) = 0$. (11)

[**25**]

3. (a) Prove that if $y_1(x)$ and $y_2(x)$ are solutions of the equation

(Eq3.1) $$y'' + P(x)y' + Q(x)y = 0$$

then

(Eq3.2) $$y_3(x) = C_1 y_1(x) + C_2 y_2(x)$$

is also a solution of this equation, for any choice of constants C_1 and C_2. (7)

(b) Suppose that y_1 and y_2 are linearly independent solutions of the homogeneous equation (Eq3.1). Assume (without proof) that *any* solution of equation (Eq3.1) can be written in the form (Eq3.2) for some choice of constants C_1 and C_2. Suppose also that y_* is a solution of equation

(Eq3.3) $$y'' + P(x)y' + Q(x)y = R(x).$$

Prove that any solution of (Eq3.3) can be written in the form

$$y = y_* + C_1 y_1 + C_2 y_2$$

for some choice of constants C_1 and C_2. (9)

(c) Given that functions $u = x\mathrm{e}^x$, $v = x\mathrm{e}^x + x$ and $w = x\mathrm{e}^x + x^2$ are solutions of an equation of the form (Eq3.3), find the functions $P(x)$, $Q(x)$ and $R(x)$ identifying that equation. (9)

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4. (a) Consider the Laplace equation in polar coordinates (ρ, θ) in the unit disk $(0 \le \rho \le 1)$,

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial u}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 u}{\partial \theta^2} = 0, \quad |u(0,\theta)| < \infty,$$

and take without proof that its general solution is

$$u(\rho,\theta) = a_0 + \sum_{m=1}^{\infty} \left(a_m \cos(m\theta) + b_m \sin(m\theta)\right) \rho^m.$$

Show that for $u(1,\theta) = f(\theta)$ the coefficients in this series are

$$a_0 = \int_0^{2\pi} \frac{f(\theta)}{2\pi}\,\mathrm{d}\theta,\ a_m = \int_0^{2\pi} \frac{f(\theta)\cos(m\theta)}{\pi}\,\mathrm{d}\theta,\ b_m = \int_0^{2\pi} \frac{f(\theta)\sin(m\theta)}{\pi}\,\mathrm{d}\theta. \tag{5}$$

(b) Consider the heat equation in a unit disk

$$\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial v}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 v}{\partial \theta^2}, \quad |v(0,\theta,t)| < \infty,\ v(1,\theta,t) = 0,$$

and take without proof that its general solution is

$$v(\rho,\theta,t) = \sum_{m=0}^{\infty}\sum_{n=1}^{\infty} \left(A_{m,n}\cos(m\theta) + B_{m,n}\sin(m\theta)\right) J_m\left(j_{m,n}\rho\right)\, \mathrm{e}^{-j_{m,n}^2 t},$$

where $J_m(z)$ is the Bessel function of the first kind of index m, and $j_{m,n}$ is its n-th positive root, $J_m(j_{m,n}) = 0$.
Derive expressions for the constants $A_{m,n}$, $B_{m,n}$ for the initial condition of the form

$$v(\rho,\theta,0) = g(\rho,\theta),$$

in terms of integrals of the initial data $g(\rho,\theta)$. (You are not requested to simplify these expressions). (10)

(c) Use results of parts (a) and (b) to write down the solution for the problem

$$\frac{\partial w}{\partial t} = \frac{\partial^2 w}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial w}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 w}{\partial \theta^2},$$
$$|w(0,\theta,t)| < \infty,\ w(1,\theta,t) = h(\theta),\ w(\rho,\theta,0) = 0,$$

for $h(\theta) = \cos\theta$, in terms of an infinite series involving Bessel functions, their roots and definite integrals with them, which you are not required to evaluate. (10)

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